

Kostiantyn Pysanyi

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PROFESSIONAL SUMMARY

Machine Learning and Computer Vision Engineer experienced in designing, training, and deploying models for complex scientific image data. Strong research and engineering background, with experience in efficient model training and data-intensive scientific applications.

TECHNICAL SKILLS

ML & CV: PyTorch/Lightning, TensorFlow, Detectron2, MMDetection, Alumentations, Scikit-learn, **OpenCV**, NumPy, Pandas, CVAT
Experimentation & Tracking: Jupyter Notebook/JupyterLab, MLflow, Tensorboard, Optuna, DVC
Infrastructure & Deployment: Linux, Docker, Kubernetes, Prefect, CI/CD (**GitLab**, Bitbucket), AWS, Triton/TensorRT
Backend & Dev Tools: Flask, REST, FastAPI, OpenAPI, FHIR, **Git**, SQL, MQTT
Languages: Python (Professional), C++17, TypeScript (Proficient), Java, MATLAB (Basic)

EXPERIENCE

Machine Learning & Computer Vision Engineer

June 2023 – July 2026

RAYLYTIC GmbH

Full-time

- Delivered production X-ray **segmentation, detection, and classification** models; improved mAP, Dice, and ROC by 20–50% over baseline and reduced training time eightfold
- Developed **3D instance segmentation and registration** for volumetric light-sheet microscopy images in a liver cancer research project
- Built and deployed a GPU-accelerated **2D–3D registration** pipeline with mesh processing and alignment for implant pose estimation from X-rays; improved acceptance by 45% and reduced error by 15%
- Implemented an MLOps pipeline with DVC data versioning, MLflow experiment tracking, and automated model evaluation
- Migrated segmentation model serving to a centralized Triton Inference Server–based REST API, cutting latency by 50%

Freelance Computer Vision Engineer

Feb 2026 – Mar 2026

Smoothr.de / Sterlix GmbH

Contract

- Developed depth-map–based camera obstruction detection for a real-time retail anti-theft system

Research Engineer

January 2022 – March 2023

Data Science Research Group

Part-time

- Developed an ML bio-signal classifier using a **high-performance computing (HPC) cluster** for pain assessment in non-responsive surgical scenarios
- Built data pipelines and training workflows for an industry ticket-routing classifier, streamlining dataset updates and model evaluation

Internships

- ML Thesis Student at RAYLYTIC GmbH – (Oct 2022 – Feb 2023): Developed an **anatomical landmark localization** model
- ML Intern at Prudsys GmbH – (Dec 2021 – Feb 2022): Automated analytic workflows and data preprocessing pipelines
- Autonomous Driving Intern at IAV GmbH – (Sep 2021 – Nov 2021): Investigated inverse reinforcement learning for self-driving
- Student Research Assistant at WHZ – (Jan 2020 – Dec 2021): Developed **semantic segmentation** for 3D lung CT tissue-density analysis; built an autoML library and predicted patient ICU admission during COVID

EDUCATION

B.Sc. Data Science with distinction (1.1 / 1.0)

University of Applied Sciences Zwickau

Oct 2019 – Apr 2023

Zwickau, Germany

M.Sc. Civil Engineering (wGPA: 90.12/100)

Odesa State Academy of Civil Engineering and Architecture

Sep 2014 – Jun 2020

Odesa, Ukraine

PUBLICATIONS & AWARDS

EFORT 2026: Validation of a 2D-3D Image Registration Method for TKA Implant Pose Estimation. Pysanyi et al.

DWG 2023: Automatic determination of lumbar segmental lordosis based on an AI algorithm. Pysanyi et al.

1st Place: National Strength of Materials Olympiad (Ukraine, 2018)

LANGUAGES

German (C1, TestDaF), English (C2, EF SET Certificate)